MILITARY SPECIFICATION

STEEL SHEET, HOT-DIP, ZINC-COATED (GALVANIZED), MINIMIZED SPANGLE, FOR AUTOMOTIVE VEHICLE BODY USE

1. SCOPE

1.1 This specification covers hot-dipped galvanized steel sheet of minimized spangle (minimal size of zinc crystals in the coating) manufactured in coils and cut lengths from continuous production lines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-698	Steel, Sheet and Strip, Low Carbon
TT-E-529	Enamel, Alkyd, Semi-Gloss
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging

Military

MIL-P-1757	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-P-15328	Primer (Wash), Pretreatment, Blue (Formula No. 117-B For Metals)
MIL-P-46105	Primer Coating, Weld-Through, Zinc-Rich

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products for Shipment and Storage
MIL-STD-171	Finishing of Metal and Wood Surfaces

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A90 Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles, Tests For

B117 Salt Spray (Fog) Testing, Method Of

B487 Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Method of Measuring

B499 Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals, Methods for Measurement Of

B529 Coating Thicknesses by the Eddy-Current Test Method: Nonconductive Coatings on Nonmagnetic Basis Metal

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Basis metal. The base metal shall be low carbon steel, conforming to specification QQ-S-698, cold-rolled and annealed.

3.1.2 Coatings. The materials used will be such as to produce galvanized coatings which meet the requirements of this specification. The molten zinc used in the hot-dip process shall be not less than 98% purity.

3.2 Construction. The zinc coating shall be applied directly to the sheet steel by immersing it into a molten zinc bath. On withdrawing the sheet from the zinc bath, treatment with steam or air will be permitted to effect minimized crystal size in the zinc coating (minimized spangle). Rolling to improve smoothness of the coating will also be permitted. The coating, immediately prior to solidification, shall not be subjected to mechanical wiping or scraping to achieve smoothness.

3.3 Coating thickness. Unless otherwise specified, the thickness of the zinc coating, on each side of ~~each test specimen~~ THE STEEL SHEET shall average 0.67 mil (17 microns).

3.4 Performance characteristics.

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3.4.1 Corrosion resistance. - Zinc-coated steel panels shall show no basis metal corrosion products when subjected to continuous exposure to salt spray for 76 hours. The presence of iron rust, as reddish-orange spots or streaks, observable to the unaided eye at normal reading distance shall be cause for rejection. Diffuse yellow staining may result from corrosion of a zinc-iron diffusion alloy and shall not be cause for rejection. Basis metal corrosion products (rust spots) within 0.5 cm from top and side edges or 1.0 cm from bottom edge shall not be cause for rejection.

3.4.2 Adhesion of galvanized coating. The adhesion of the galvanized (zinc, zinc-iron diffusion layer) coating to the sheet steel shall be such that, after application of the Bend Test there shall be no separation of the coating from the basis metal or flaking or spalling of zinc from underlying coating.

3.4.3 Paint adhesion. - The adhesion of paint system applied to galvanized steel specimens shall be such that no separation is visible to the unaided eye upon application of the Tape Test.

3.4.4 Weldability. - The condition and uniformity of the zinc coating shall be such as to permit good quality spot or seam resistance or gas-metal arc welding when employing generally accepted or existing good welding practices.

3.5 Dimensions. - A single standard mill-production coil shall be a galvanized sheet of 250 ft (76 M) continuous length. The length of the galvanized sheet (Standard mill-production or fraction thereof) or the width shall not be less than that ordered.

3.6 Workmanship.

3.6.1 Basis metal. - The basis metal shall be free from defects that will mar the appearance of the finished products. It shall be subjected to such cleaning, pickling or fluxing procedures as are necessary to yield coatings as hereinafter specified.

3.6.2 Zinc coating. - The galvanized sheets shall be of prime finish, free from apparent surface defects. The zinc coating shall be smooth, adherent, free from pits, blisters, nodules, lumps and accentuated spangles that are likely to be apparent or detectable after painting with the specified finishing system. When visible, spangles will be accepted as minimum if the zinc crystals are of small size, i.e., not more than 1/8" (3 mm) average diameter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilized his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Samples for test. - The test specimens for coating thickness determinations, and as may be required for coating weight/area determinations, shall be taken from a sample piece, approximately 12 in (30 cm) in length by the as-coated coil width, cut from the free end of a coil or from a cut sheet from a coated coil. Two specimens shall be taken from the middle area of the sample piece, and one from each side, not closer than 4 in (10 cm) from the side edge. Test specimens for other determinations shall be taken from additional cut sheets, to within 10 cm from the side edge, or may be taken from remaining portions of sample pieces. Unless otherwise specified, samples for test specimen dimension and inspection shall be accordance with Table I.

TABLE I

Test	Sample Size	Specimen Size	Requirement Paragraph	Method Paragraph
Thickness	4	4 X 6" (10.2 X 15.3cm)	3.3	4.3.2
Salt Spray	8	4 X 6" (10.2 X 15.3cm)	3.4.1	4.3.3
Zinc Coating Adhesion	4	4 X 12" (10.2 X 30.6cm)	3.4.2	4.3.4
Paint Adhesion	4	4 X 6" (10.2 X 15.3cm)	3.4.3	4.3.5
Welding	6	4 X 12" (10.2 X 30.6cm)	3.4.4	4.3.6

Additional specimens may be prepared and held in readiness for retesting, should this become necessary for any of the requirements.

4.2.2 Noncompliance. - Any lot of galvanized sheet shall be accepted or rejected on the basis of Table I inspection. In the event of rejection acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After corrective action has been taken Table I inspection shall be repeated on sample units. Final acceptance shall be withheld until reinspection illustrates corrective action has been successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the contracting officer.

4.2.3 Inspection of preparation for delivery. - Sample packages and packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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4.3 Methods of test and examination.

4.3.1 Surface examination. - The panel specimens shall be examined for compliance with materials, workmanship and physical dimensions of rolled sheets. (See 3.5, 3.6.1 and 3.6.2). Failure to pass such examination shall be cause for rejection. After chemical or solvent cleaning (see 6.3.2, 6.3.3 and 6.3.4), the specimen surface shall be smooth and free of irregularities, such as: crystals in relief, nodules, lumps, pits and gouges, as determined visually and by drawing the fingertips, with slight pressure, over the surface.

4.3.2 Thickness. - The preferred methods of testing coating thickness shall be the Magnetic Gauge Method in accordance with ASTM B499 and the Eddy-Current Method in accordance with ASTM B529. A measurement is made at five points on each side of the panel specimen not closer than 1.6 in (ca + cm) from each other and not closer than 0.6 in (ca 1.5cm) from the edges. A minimum of three separate readings shall be made at each point and averaged. The thickness at each of the five points shall be averaged to give the coating thickness. Should any side among the group of four specimens fail to comply with the requirements of para 3.3., a second group of samples from the same coil or sheet piece shall be measured for coating thickness. Should one or more of the sides of the second set of four panels fail to comply with the thickness requirements, the same panels shall be stripped of coating in a hydrochloric acid solution (see 6.3.1). One side shall be masked during initial stripping of each panel so that the weight of the coating for each side may be ascertained. The weight loss per side shall be at least 3.8 g. Should any side among the specimens fail to comply with this requirement, the lot of galvanized sheet shall be rejected. The chemical method in accordance with ASTM A90 and the microscopic method in accordance with B487 will also be acceptable methods of testing coating thickness (see 6.2).

4.3.3 Corrosion resistance (Salt spray test). - The salt spray test shall be performed (representing different areas and surfaces of the galvanized sheet) in accordance with ASTM Standard B117 and Table I. The surface of the specimens shall be bare and clean. Grease films shall be removed by solvent or solvent vapor (6.3.2) and chemical films (e.g., chromate or phosphate treatment) shall be removed by chemical means (6.3.3). If necessary to assure removal of fine dust or adherent particles, specimens may be wiped with a cloth wet with solvent (6.3.4). Four specimens in each of two groups shall be tested; in one group one side of the original sheet shall be exposed, in the other group the other side.

4.3.4 Adhesion of zinc coating (Bend test). - Four panel specimens taken from the sample piece shall be bent through 180°, perferably with the aid of a smooth-jaw vise. The bent portion shall be placed in the vise and squeezed to reduce the radius of the bend until the legs, just beyond the bend, are apart by approximately four times (or less) the panel thickness. The outside surface of the bend is examined for evidence of separation of the zinc coating from the basis steel. Examination for separation or delamination of the zinc coating shall be performed at 7X. The lifting or flaking of the zinc which extends more than 1/4 in (0.65 cm) from the edge of the specimen, at the bend, shall be cause for rejection.

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4.3.5 Paint adhesion (Tape test). - Four panel specimens shall be cleaned and then painted using a wash primer (MIL-P-15328), a chromate primer (MIL-P-1757), and an enamel top coat (TT-E-529) as described in 6.3.5 and in accordance with the recommended thicknesses of MIL-STD-171B. Conventional vapor degreasing of oiled galvanized sheet shall be sufficient preparation for painting so as to conform to paint adhesion requirements. Should grit or other soil not be throughly removed by degreasing, wiping with a dry cloth or a cloth wet with solvent may be necessary. The paint system shall be allowed to dry for 24 hours. On two separate areas, on each specimen, a 2 in (5 cm) long strip of water-resistant pressure sensitive tape, approximately 3/4 in (1.9 cm), conforming to Fed Spec PPP-T-60, Type I or III, Class 1 or 2, shall be pressed firmly onto the painted surface taking care to eliminate all air bubbles under the tape. After at least 10 seconds, grasp a free end of the pressed tape (made available by either previously turning one end of the tape on itself on the adhesive side or using a tape length long enough to extend beyond the edge of the panel specimen) and by rapid motion strip it from the surface by pulling the tape back on itself at an angle of 180°. Observe the surface of the specimens uncovered by stripping off the tape, and the tape itself for evidence of removal or peeling of the paint system. Flecks of paint on the tape shall be disregarded if metal underlying the paint on the specimen is not exposed.

4.3.6 Weldability. - Pairs of panel specimens (See Table I) shall be overlapped about 3/4 in (2 cm) along the larger edge, and held thus for welding. Prior to readying the panels shall have been cleaned. Resistance spot welds shall be accomplished at one 1 in (2.5 cm) intervals along the overlap on two pairs of specimens. Separately gas metal arc and shielded metal arc bead welds shall be accomplished on one side of each of two pairs of bare, clean specimens. The requirement to perform weld tests on sheets coated with weld-through primer, MIL-P-46105 shall be accomplished when required by the agency procuring the items fabricated from the galvanized sheet. Good weldability shall be based on any or all of the following requirements:

a. Resistance spot welding - The shear strength of a spot weld, of approximately 0.24 in) diameter, shall be at least 5950 lbs (2700 kg). When required by in procuring agency, it shall be demonstrated, with the galvanized sheet under test and using industrial, production welding process, that 2000 consecutive spot welds of the indicated quality can be effected, without requiring changing or redressing of the welding electrodes. A typical industrial schedule capable of fulfilling the specified requirements is presented in 6.3.6.

b. Gas metal arc welding, by short circuiting drop transfer or by automatic setup with CO_2 shielding - A typical industrial method is given 6.3.7. Sound, lap-fillet welding of the galvanized sheet shall be accomplished by this method. The weld bead shall be even, and the zinc coating on the back of the underlying galvanized sheet steel which may be backed with metal plate shall not be penetrated or separated on completion of the weld operation. A typical industrial process capable of yielding these requirements is presented in 6.3.7.

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c. Shielded metal arc welding - Sound, lap-fillet welding of the galvanized steel shall be accomplished by this method. The weld bead shall be even, and the zinc coating on the back of the underlying sheet, which may be backed with metal plate, shall not be penetrated or broken after completing the welding. A typical industrial process which will meet these requirements is described in 6.3.8.

Excessive metal expulsion during welding or sticking of the welding electrodes to the work which is detrimental to effecting good quality welds and which cannot be rectified by adjusting the process, shall be cause for rejection of the galvanized sheet being tested.

4.3.7 Chemical Analysis. - As may be required, chemical analysis of the basis steel sheet shall be performed or supplied for conformance to Federal Specification QQ-S-698 (see 6.2). Also as may be required, analysis of zinc, sampled from the hot-dip tank or the coating on the galvanized steel sheet, shall be in conformance to QQ-S-698 for base metal composition (see 6.2).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2)

5.1.1 Level A. - Steel sheet, zinc-coated, shall be preserved and packaged in accordance with level A requirements of MIL-STD-163.

5.1.2 Level C. - Preservation and packaging shall be in accordance with contractor's commercial practice. The galvanized sheet may be supplementally treated or oil-coated to provide corrosion prevention during shipment and storage.

5.2 Packing.- Packing shall be level A or C as specified. (See 6.2)

5.2.1 Level A. - Steel sheet, zinc-coated, shall be packed in accordance with the level A requirements of MIL-STD-163.

5.2.2 Level C. - Steel sheet, zinc-coated, shall be packed in such a manner so as to insure carrier acceptance and safe delivery at destination in containers complying to the rules and regulations applicable to the mode of transportation.

5.3 Marking. - Items and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The minimum spangle galvanized steel sheets are intended for use in fabricating automotive vehicle bodies and other applications which require painting and not reveal the crystaline nature of the zinc-coating.

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6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether supplemental treatment for corrosion prevention during shipment and storage is desired.

c. Whether chemical analysis for base metal composition in accordance with QQ-S-698 is desired.

d. Levels of preservation, packaging, packing and applicable marking required.

e. Specific method of testing coating thickness if other than the Magnetic Gauge or Eddy-Current Methods.

6.3 General information.

6.3.1 Stripping of the galvanized coating. - Stripping of the galvanized coating may be accomplished by immersing the specimen in a hydrochloric acid aqueous solution consisting of 50 parts by volume of 1.19 sp. gr., HCl. Approximately two (2) minutes immersion time will accomplish complete stripping. One liter of solution will effectively strip up to two square feet of galvanized surface. Throughout the stripping operation solution temperature should be maintained between 20 to 30°C (68 to 86°F).

6.3.2 Degreasing of specimens. - Solvent degreasing may be accomplished by immersing the specimens in mineral spirits (Stoddard solvent) or trichloroethylene. If the film is light, acetone may be used followed by rinsing with dry acetone or 95 percent methyl or ethyl alcohol. Vapor degreasing with stabilized or inhibited trichloroethylene is more convenient, if facilities are available; it is recommended for pieces of galvanized sheet.

6.3.3 Removal of chromate film. - Chromate film may be removed by immersing the chromated galvanized specimen in sodium carbonate solution, 10 percent $Na_2 CO_3$ by weight, at ca 66°C (150°F), for 3 to 4 minutes. Following thus the specimens are rinsed and the gelantinous film is removed by rubbing with a soft cloth or sponge and a fine abrasive (e.g., nonbleaching household scouring powder) washed clean with water, and rinsed with dry acetone, or in 95 percent methyl or ethyl alcohol.

Phosphate film may be removed by immersing the phosphated, galvanized specimen in ammonium hydroxide solution (28 percent NH_3), at room temperature for five minutes. Rinse on removal from the solution and rub gently with a soft cloth or sponge and a fine abrasive (e. g., nonbleaching household scouring powder or followed by washing with warm water, then rinsing in dry acetone, or in 95 percent methyl or ethyl alcohol.

6.3.4 Cleaning with dry acetone or alcohol. - A clean cheese cloth or gauze wad moistened with dry acetone or alcohol may be used to rub away the more tenacious dust or smut not removed by the degreasing operation.

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6.3.5 Thickness of test finishes. - The paints and coating thickness of the separate components of the test finishing system comply with the recommendations of MIL-STD-171 and are as follows:

a. Wash primer, MIL-P-15328 - shall be applied to yield a dry film thickness of 0.5 ± 0.1 mil (0.013 ± 0.0025 mm).

b. Primer, MIL-P-8585 - shall be applied to yield a dry film thickness of 0.5 ± 0.1 mil (0.013 ± 0.0025 mm).

c. Enamel, TT-E-529 - shall be applied to yield a dry film thickness of 1.0 ± 0.2 mil (0.025 ± 0.005 mm).

6.3.6 Resistance spot welding. - The following schedule applies to resistance spot welding, using a 150 KVA single phase welder:

Electrode	5/8 in dia. (15.9 mm)	RWMA, Cr. A, Cl 2 (M28)
Electrode Shape	1/4 in dia. (6.4 mm)	120 truncated cone
Electrode force	1000 lbs. (454 kg)	
Weld time	22 cycles	
Weld current	13200 to 14,500 amperes (rms)	
Squeeze time	20 cycles	
Hold time	50 cycles	
Weld overlap	3/4 in (19.1 mm)	
Weld spacing	1-1/8 in (28.6 mm)	
Weld diameter	0.24 in (6.1 mm) (min)	
Weld strength	2500 lbs (1134 kg) (min)	

6.3.7 Gas metal arc welding. - The following schedule applies to gas metal arc welding, using the short-circuiting drop transfer technique and an automatic set-up with CO_2 shielding and a power source consisting of a 300 ampere DC rectifier, in making a lap fillet weld a copper back-up plate is used to minimize the loss of zinc:

Current	110 to 120 amperes dc
Arc voltage	22.5 volts
Shield gas	CO
Gas flow rate	35 ft/hr (ca/m)
Electrode size	0.034 in (0.87 mm) dia.
Electrode type	AWS E 70S3 (Hobart 25)
Travel speed	30 in (76 cm)/min

6.3.8 Shielded metal arc welding. - The following schedule applies to shielded metal arc welding, using a 300 VDC rectifier, in making a lap-fillet weld. A copper back-up plate is used to minimize loss of zinc:

Current	90 amps dc
Arc voltage	22 volts dc
Electrode	ca. 1/8 in (3.2 mm) dia. E6012 (Westinghouse ZP12)
Travel speed	8 in (20.3 cm)/min (approx.)

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